Assignment # 02

Data Science Tools & Techniques (MS DS)

Deadline: 14th October 2026 @ 11:59 PM

Total Marks = 100

Instructions:

- Combine all your work in one folder. The folder must contain only deliverable files (no binaries, no exe files, etc.).
- Rename the folder as ROLL-NUM_SECTION (e.g., 22i-0001_A) and compress the folder as a zip file. (e.g. 23i-0001_A.zip). Do not submit a .rar file.
- All the submissions will be done on Google Classroom within the deadline. Submissions other than Google Classroom (e.g., email, etc.) will not be accepted.
- The student is solely responsible for checking the final zip files for issues like corrupt files, viruses in the file, and mistakenly executed sent. If we cannot download the file from Google Classroom for any reason, it will lead to zero marks in the assignment.
- Displayed output should be well-mannered and well-presented. Use appropriate comments and indentation in your source code.
- **Be prepared for a viva or anything else after the submission of the assignment.**
- If there is a syntax error in the code, zero marks will be awarded in that part of the assignment.
- Understanding the assignment is also part of the assignment.
- **Zero marks will be awarded to the students involved in plagiarism. (Copying from the internet is the easiest way to get caught.)**
- **Late Submission** will **not** be entertained, and **no request retake** will be accepted as per the course policy.

Tip: For timely completion of the assignment, start as early as possible.

Note: Follow the given instruction; failing to do so will result in a zero.

Q#1: Data Distributions and Their Applications (100 Marks)

Perform the following tasks for this question.

Dataset:

Use the following dataset for experiments:

- **Dataset #1:** Dataset1 (dataset.csv) – Provided.
- **Dataset #2:** CIFAR-10 Dataset (Download via torchvision Datasets or keras Datasets)

Task 1: Data Analysis (30 Marks):

Using Dataset #1 (dataset.csv):

- Perform data preprocessing to fix issues in the dataset:
  - ✓ Handle missing values.
  - ✓ Detect and treat outliers.
  - ✓ Remove duplicate rows.
- Perform EDA (Exploratory Data Analysis):
  - ✓ Summary statistics (mean, median, variance)
  - ✓ Distribution plots (histograms, density plots).
  - ✓ Boxplots for outlier detection.
- Identify which features follow normal, uniform, skewed, or multimodal patterns.

Task 2: Case Study with Real Data (30 Marks):

Using Dataset #2 (CIFAR-10):

- Load the dataset and explain its structure (training/test sets, classes).
- Select any three classes (e.g., airplane, cat, car).
- For each class:
  - ✓ Visualize random samples of images.
  - ✓ Extract simple numerical features (e.g., pixel intensity histograms, mean RGB values).
  - ✓ Analyze their distributions (normal, skewed, multimodal, etc.).
- Discuss whether CIFAR-10 can be considered IID or Non-IID in your experiments, and why.

Task 3: Report Writing (20 Marks):

- Submit a brief report (~500 words) covering:
  - ➢ Explanation of all distribution types with plots.
  - ➢ Key insights from these two datasets**.**
  - ➢ Distribution analysis of these two data sets' features.
  - ➢ Why understanding distributions is important in Data Science.

Deliverables

- ➢ **Jupyter Notebook** with code, outputs, and plots for all tasks.
- ➢ **Report (Word/PDF, ~500 words)** summarizing findings.